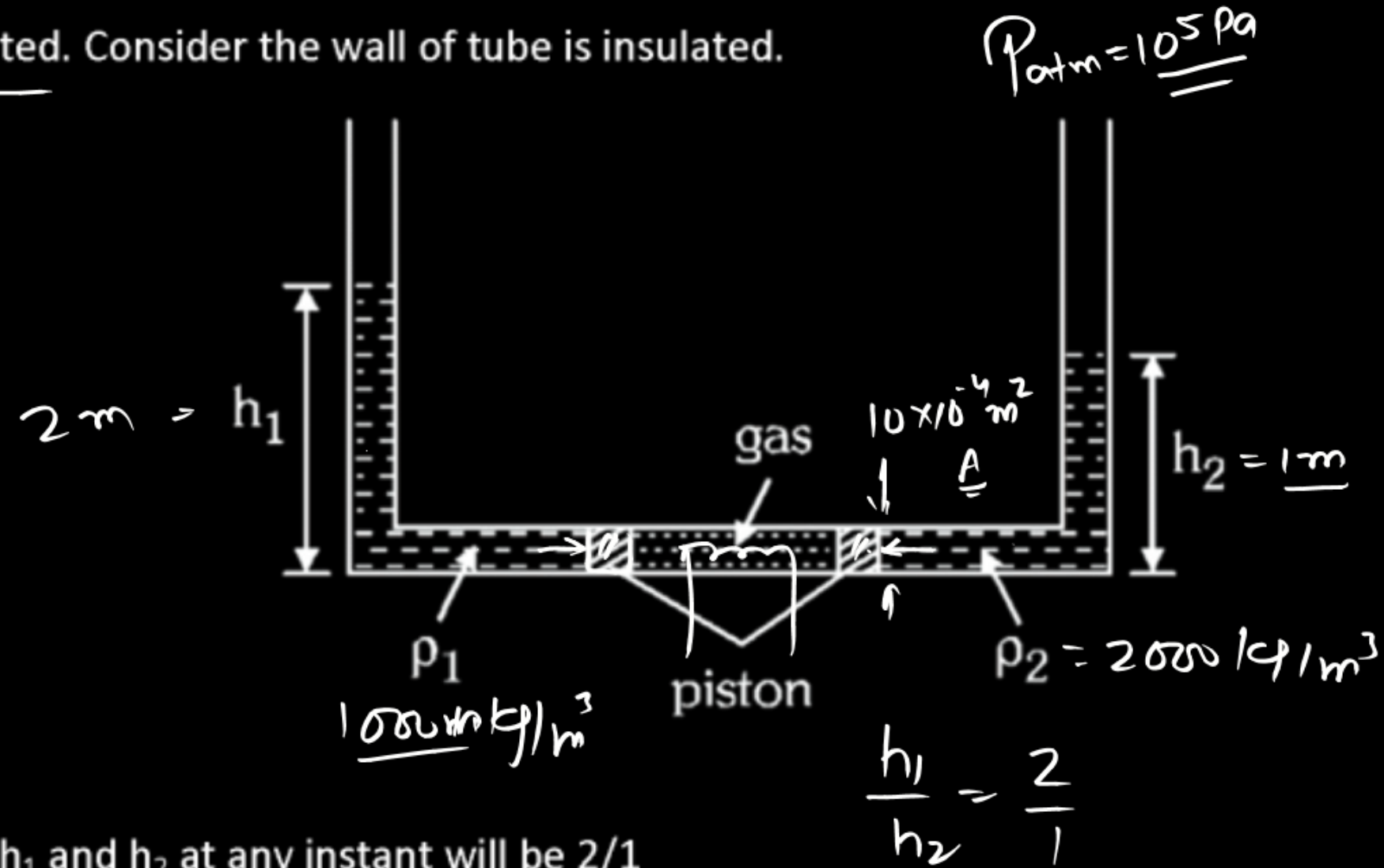


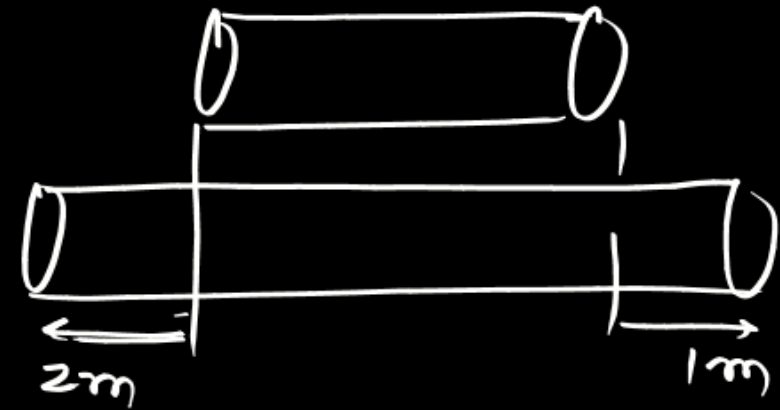
Figure shows a tube whose two limbs are vertical and one is horizontal of sufficient length. An ideal gas is enclosed between two massless non-conducting movable piston of cross-sectional area 10 cm^2 . A liquid of density $\rho_1 = 1000 \text{ kg/m}^3$ and $\rho_2 = 2000 \text{ kg/m}^3$ are poured in the vertical limbs of height of $h_1 = 2\text{m}$ and h_2 . Now gas is slowly heated. Consider the wall of tube is insulated.



$$P_{\text{atm}} + \rho_1 g h_1 = P_{\text{atm}} + \rho_2 g h_2$$

$$\rho_1 h_1 = \rho_2 h_2$$

$$\frac{\rho_1}{\rho_2} = \frac{h_2}{h_1} = \frac{1}{2}$$

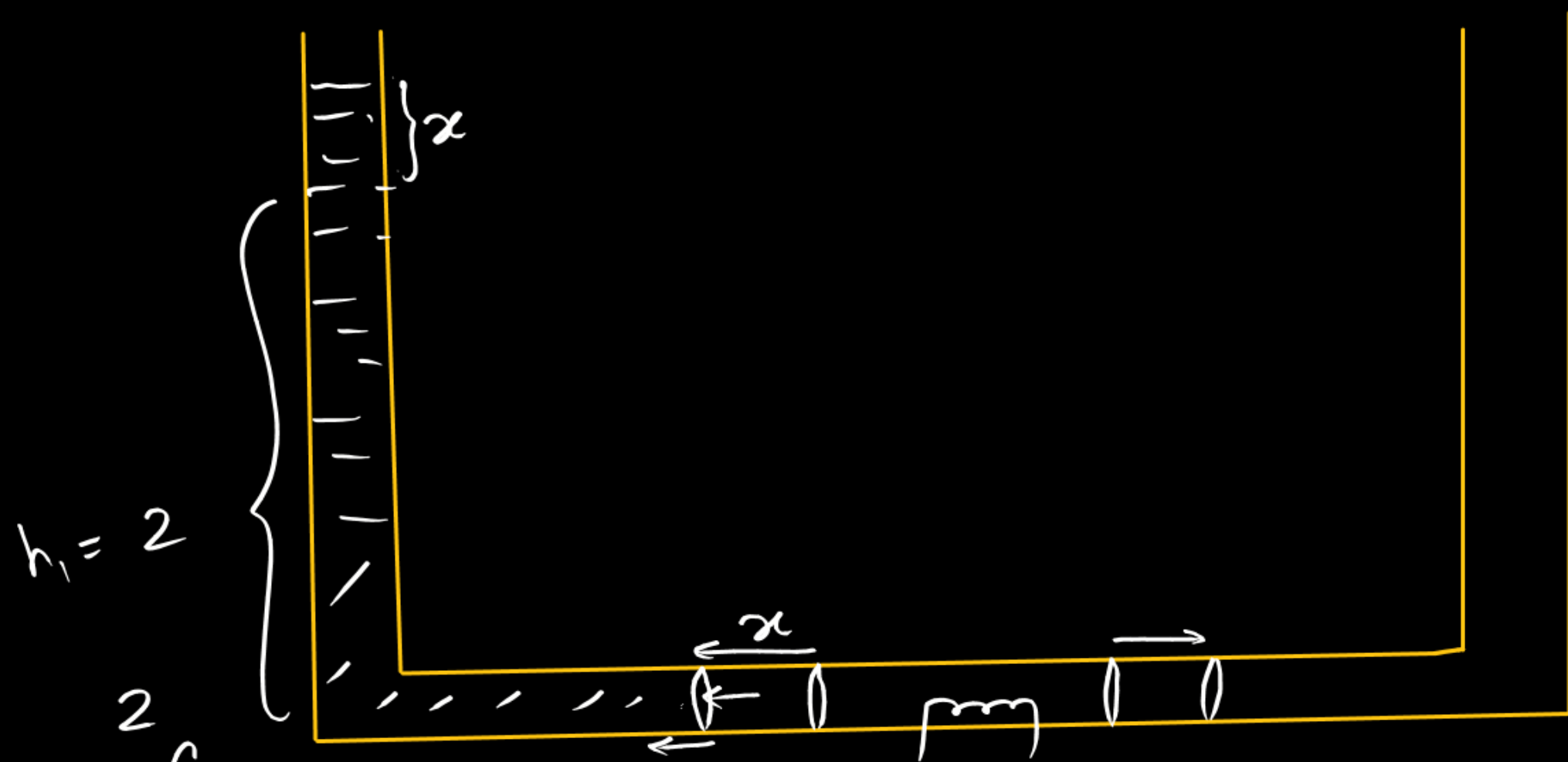


(A) Ratio of h_1 and h_2 at any instant will be $2/1$

(B) If due to heating separation between piston is increased by 3m then displacement of piston on left will be 1m

(C) If due to heating separation between piston is increased by 3m then displacement of piston on left will be 2m

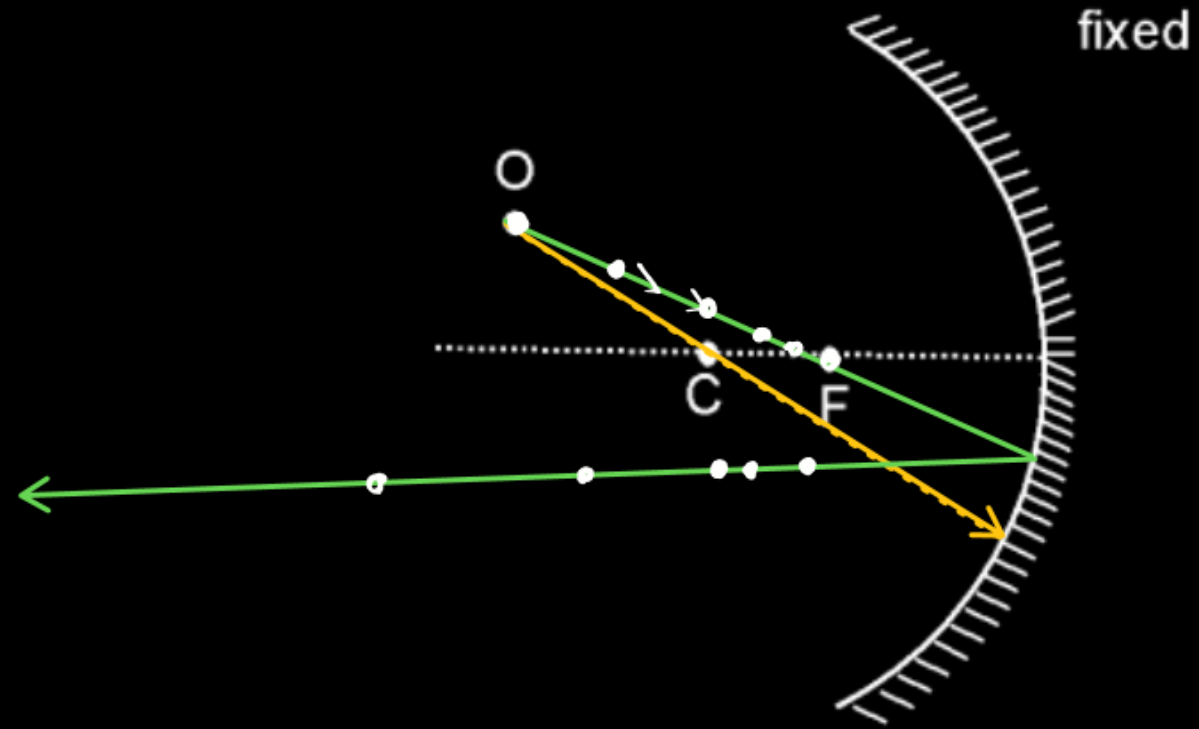
(D) The work done by gas for above situation is 390 Joule



$$\int_0^2 \left[P_{atm} + \rho_1 g [h_1 + x] \right] A dx$$

$$\left[P_{atm} A x 2 + \rho_1 g h_1 A x 2 + \frac{\rho_1 g A x^2}{2} \right] + P_{atm} A x 1 + \rho_2 g h_2 A x 1 + \frac{\rho_2 g A x 1^2}{2}$$

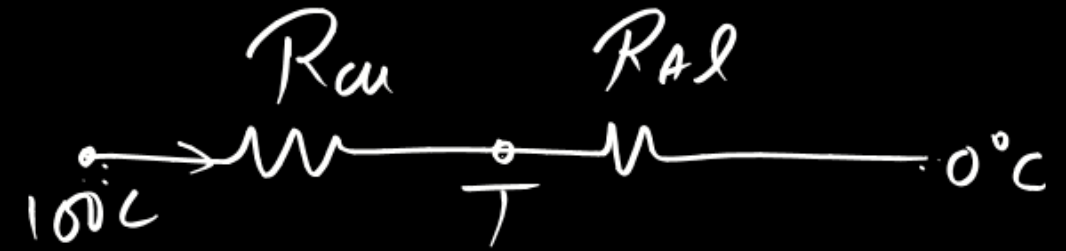
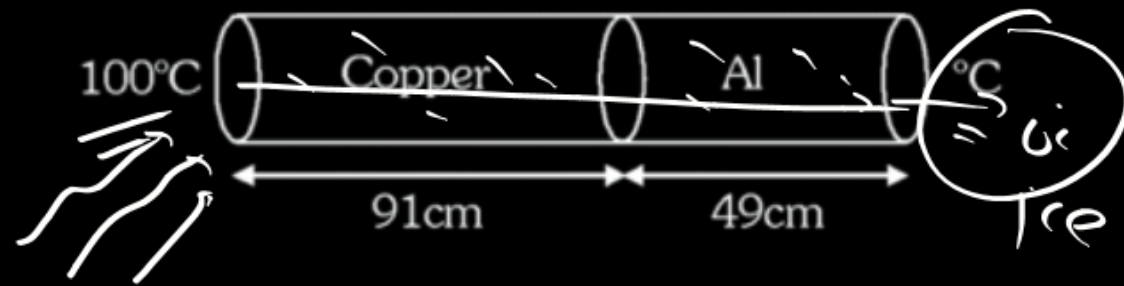
An object above principal axis is towards left of centre of curvature of mirror. Which of the following is/are correct at given instant (consider image formation by paraxial rays)



- ☒ (A) If object moves parallel to principal axis then image also must move parallel to principal axis.
- ☒ (B) If object moves perpendicular to principal axis then image also must move perpendicular to principal axis.
- ☒ (C) Object and image can move along same line.
- ☒ (D) If object moves towards the mirror such that its line of motion is passing through focus then image will move parallel to principal axis.

A bar of Copper length 91 cm and Aluminium of length 49 cm are joined together end to end. Both rod have circular cross-section with diameter $\sqrt{\frac{1}{50\pi}}$ m. Free end of Copper & Aluminium are maintained at 100°C and 0°C respectively. For maintaining temperature steam at 100°C is continuously flowing at the end of copper & ice is continuously melting at the end of Aluminium. After reaching steady state, choose correct statement(s) [Thermal conductivity of copper is $91 \text{ cal m}^{-1}\text{C}^{-1}\text{s}^{-1}$ and Thermal conductivity of Aluminium is $49 \text{ cal m}^{-1}\text{C}^{-1}\text{s}^{-1}$, $L_f = 80 \text{ cal/g}$ and $L_v = 540 \text{ cal/g}$].

Neglect any thermal resistance at the junction.



$$R_{Cu} = \frac{l}{KA} = \frac{0.91 \text{ m} \times 4 \times 50}{91 \text{ cal m}^{-1}\text{C}^{-1}\text{s}^{-1} \times \pi \times \frac{1}{50\pi} \text{ m}^2} = 2 \frac{^\circ\text{C} \times \text{sec}}{\text{cal}}$$

$$R_{Al} = \frac{0.49 \times 4 \times 50}{49 \times \pi \times \frac{1}{50\pi}} = 2 \frac{^\circ\text{C} \times \text{sec}}{\text{cal}}$$

(A) Heat flow per unit time across the junction is 25 cal/sec.

(B) The rate of melting ice is $\frac{5}{16} \text{ g/sec}$.

(C) Rate of condensation of steam is $\frac{5}{108} \text{ g/sec}$.

(D) Temperature of junction is 50°C

$$\frac{100 - T}{2} = \frac{T - 0}{2}$$

$$T = 50^\circ\text{C}$$

$$R_{\text{eq}} = 4 \frac{^{\circ}\text{C} \cdot \text{sec}}{\text{cal}}$$

100 $\xrightarrow{\quad}$ 0°C

$$Q_m = \frac{100^{\circ}\text{C} \times \text{cal}}{4^{\circ}\text{C} \times \text{sec}}$$

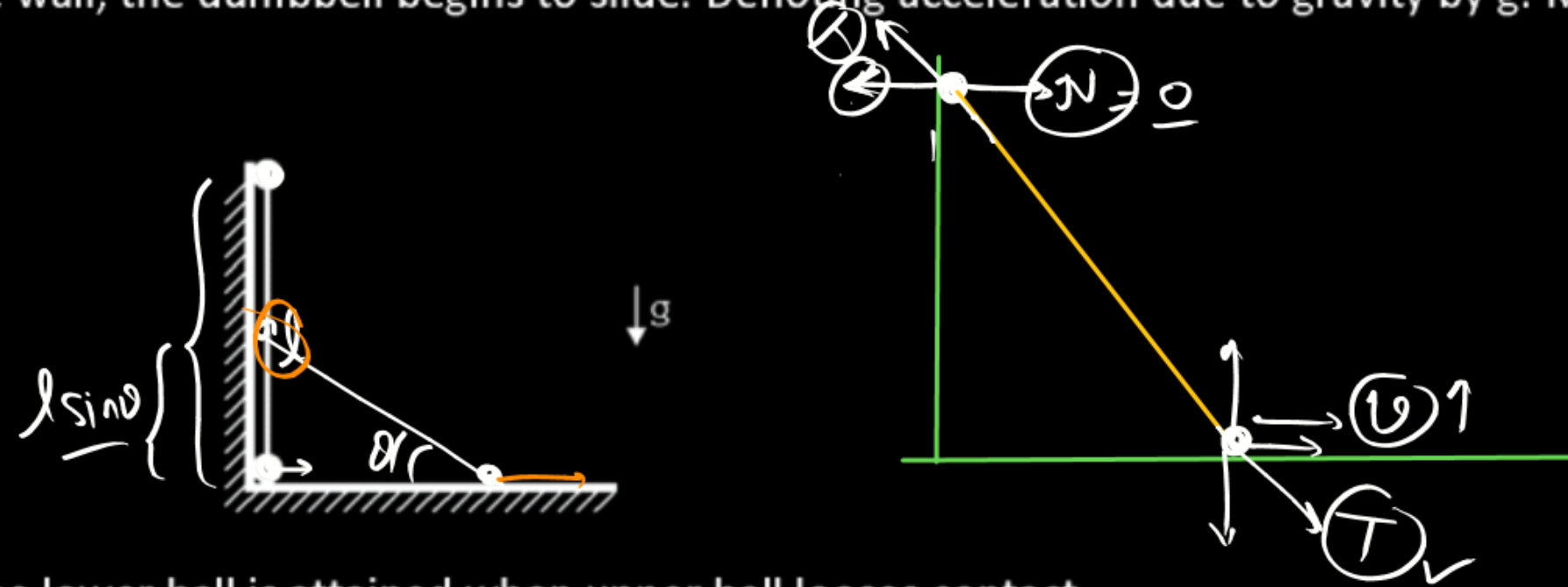
$$= 25 \frac{\text{cal}}{\text{sec}} = \frac{dQ}{dt}$$

$$\frac{dQ}{dt} = 25 \frac{\text{cal}}{\text{sec}} = \left(\frac{dm}{dt} \right) L_f = \frac{dm}{dt} \times 80 \frac{\text{cal}}{\text{gm}} = \frac{dm}{dt} \times L_v$$

$$\frac{dm}{dt} = \frac{25}{80} \frac{\text{gm}}{\text{sec}} = \frac{5}{16} \frac{\text{g}}{\text{sec}}$$

$$\frac{dm}{dt} = \frac{25 \text{ cal/sec}}{540 \text{ cal/gm}} = \frac{5}{108} \frac{\text{gm}}{\text{sec}}$$

A dumbbell constructed by affixing small identical balls each of mass m at the ends of a light rod of length ℓ stands vertically on a frictionless floor touching a frictionless wall as shown in the figure. If the lower ball is gently pushed away from the wall, the dumbbell begins to slide. Denoting acceleration due to gravity by g . Mark the **CORRECT** statements:



☒ (A) The maximum speed of the lower ball is attained when upper ball loses contact.

☒ (B) The maximum speed of the lower ball is attained when tension in the light rod is zero.

☒ (C) The maximum speed of the lower ball is $\sqrt{\frac{8gl}{27}}$

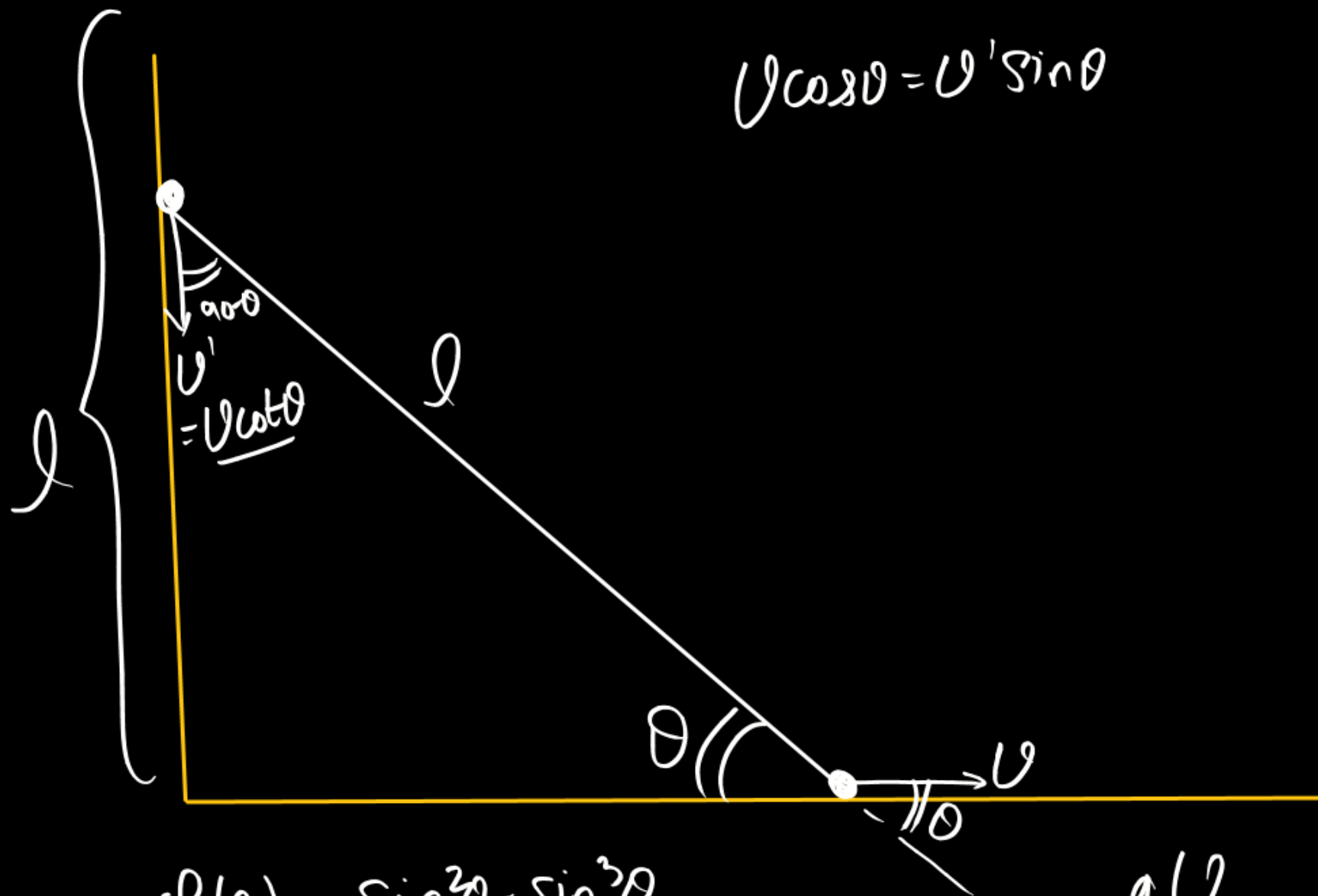
$v \rightarrow \max$

$$\frac{dv}{dt} = 0 = a \quad \underline{f=0}$$

☒ (D) The speed of the upper ball when lower ball acquires maximum speed is $\sqrt{\frac{10gl}{27}}$

$$\frac{3}{\sqrt{5}} \times \frac{10}{\sqrt{5}} = 2$$

$$v' = v \cos \theta = \sqrt{\frac{8gl}{27}} \times \frac{\sqrt{5}}{2} = \sqrt{\frac{28gl \times 5}{27 \times 4}} = \sqrt{\frac{10gl}{27}}$$



$$v \cos \theta = v' \sin \theta$$

MEC

$$mgl(1 - \sin \theta) = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 \cos^2 \theta$$

$$v^2 = \frac{2gl(1 - \sin \theta)}{\cos^2 \theta}$$

$$= (2gl) \sin^2 \theta (1 - \sin \theta)$$

$$\theta = 90^\circ$$

$$v = \infty$$

$$\theta = 0$$

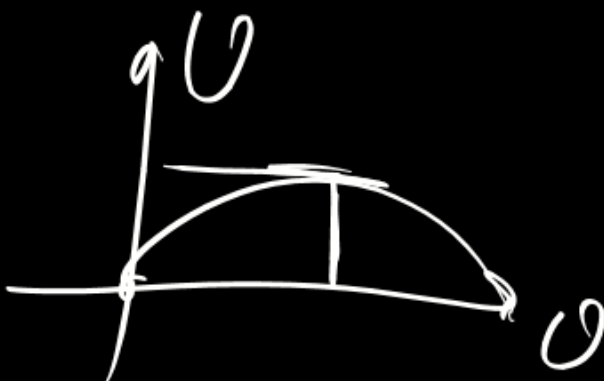
$$v_{\max}^2 = 2gl \times \frac{4}{9} \times \frac{1}{3} = \frac{8gl}{27}$$

$$v_{\max} = \sqrt{\frac{8gl}{27}}$$

$$f(\theta) = \sin^2 \theta - \sin^3 \theta$$

$$f'(\theta) = 2 \sin \theta \cos \theta - 3 \sin^2 \theta \cos \theta = 0$$

$$\sin \theta = \frac{2}{3}$$



An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P_1 . Now a transparent cylinder which does not absorb any light of a refractive index $\mu(= \sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

Case-1 : Source is kept on the axis of the cylinder, just outside the cylinder.

Case-2 : Source is kept on the axis of the cylinder, just inside the cylinder.

In case-1 power crossing the circular hole is P_2 and in case-2 it is P_3 .

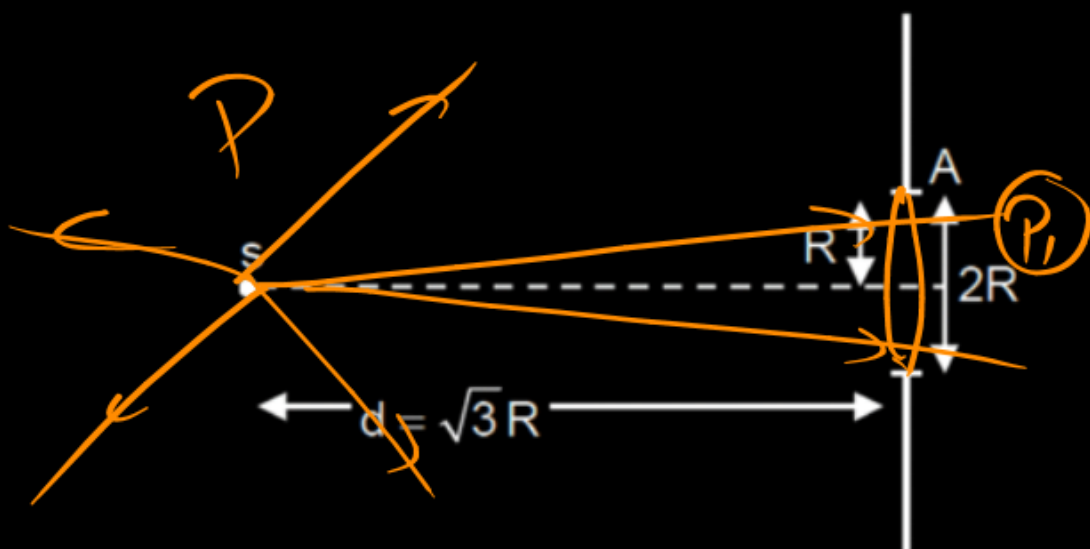


Figure -1

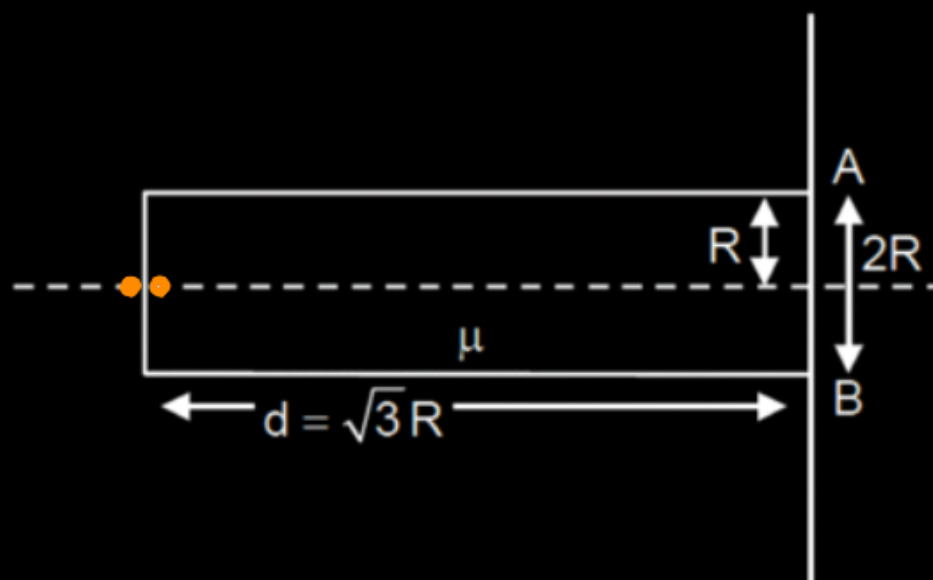


Figure -2

Answer the following two questions :

Value of P_2 is :

(A) $\frac{2P_1}{2 - \sqrt{3}}$

(B) $\frac{P_1}{2 - \sqrt{3}}$

(C) $\frac{2P_1}{2 + \sqrt{3}}$

(D) $\frac{P_1}{2 + \sqrt{3}}$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P_1 . Now a transparent cylinder which does not absorb any light of a refractive index $\mu(= \sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

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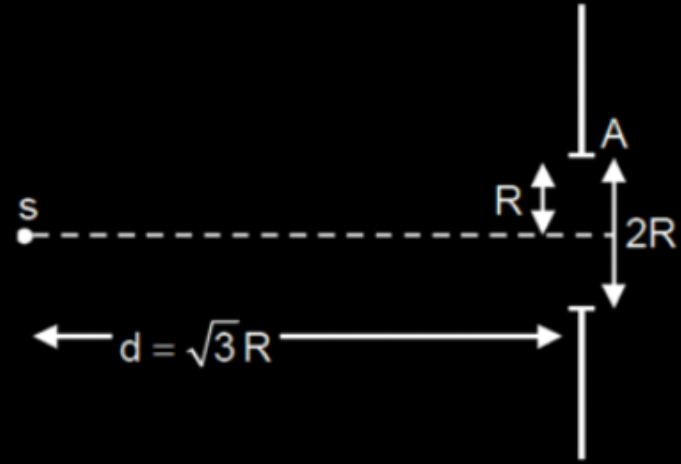


Figure -1

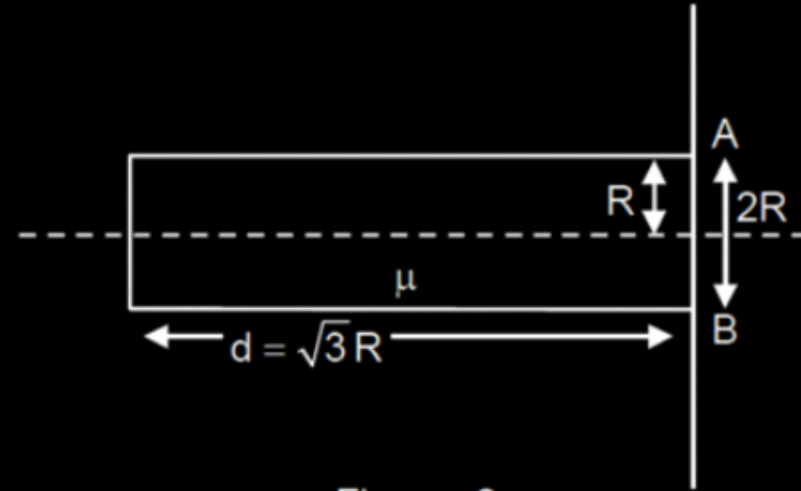


Figure -2

Answer the following two questions :

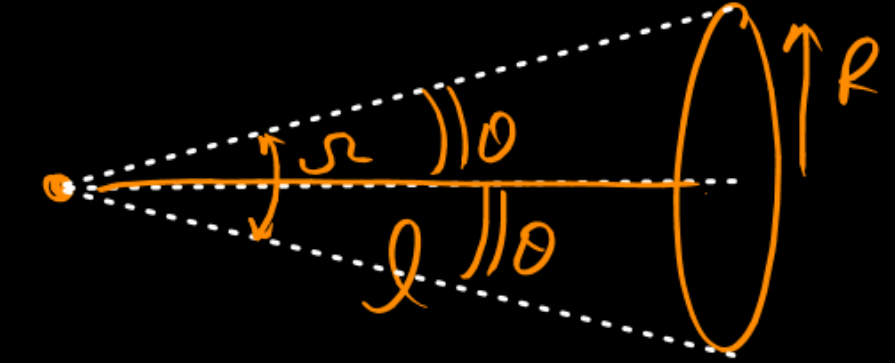
Value of P_3 is :

(A) $P_1 \left(\frac{2 + \sqrt{2}}{2 + \sqrt{3}} \right)$

(B) $P_1 \left(\frac{2 + \sqrt{2}}{2 - \sqrt{3}} \right)$

(C) $P_1 \left(\frac{2 - \sqrt{2}}{2 + \sqrt{3}} \right)$

(D) $P_1 \left(\frac{2 - \sqrt{2}}{2 - \sqrt{3}} \right)$



$$\Omega = 2\pi(1 - \cos\theta)$$

$$= 2\pi \left\{ 1 - \frac{d}{\sqrt{d^2 + R^2}} \right\}$$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P_1 . Now a transparent cylinder which does not absorb any light of a refractive index $\mu(= \sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

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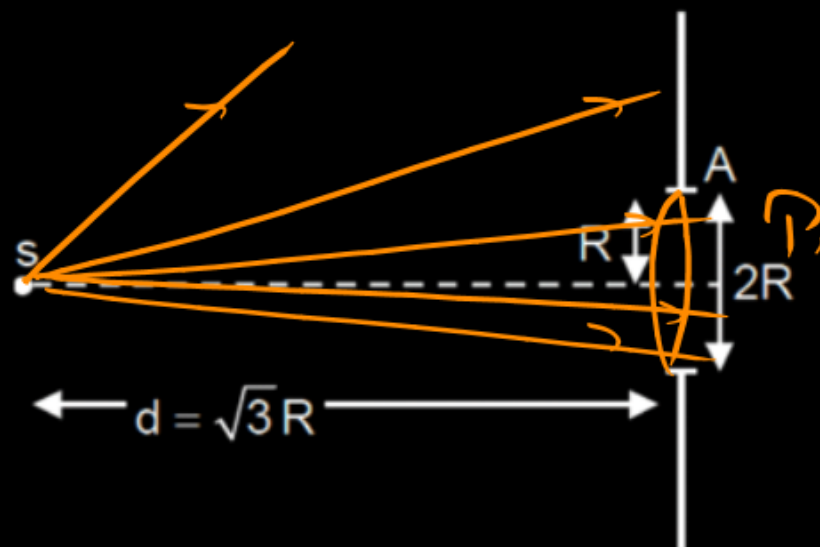


Figure -1

Answer the following two questions :

Value of P_2 is :

(A) $\frac{2P_1}{2-\sqrt{3}}$

(B) $\frac{P_1}{2-\sqrt{3}}$

(C) $\frac{2P_1}{2+\sqrt{3}}$

(D) $\frac{P_1}{2+\sqrt{3}}$

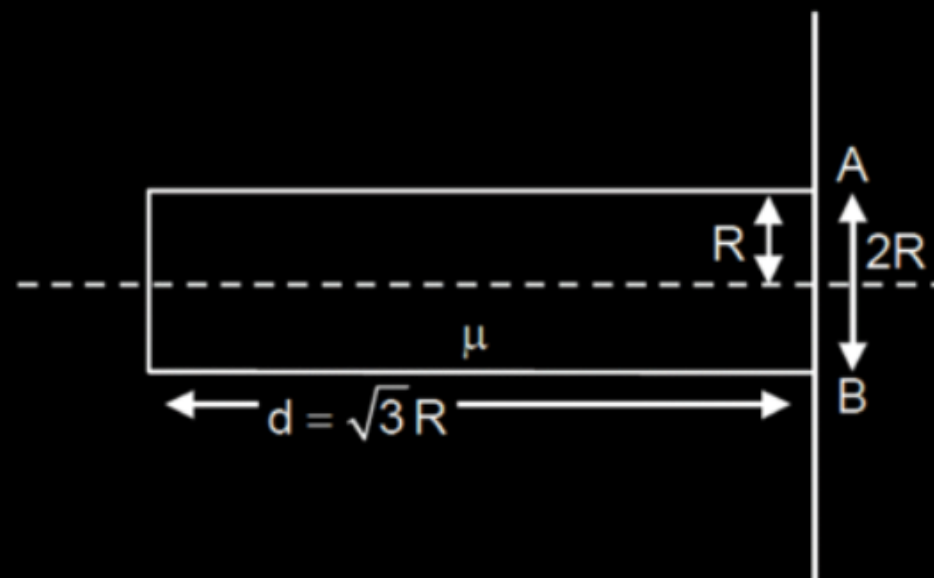
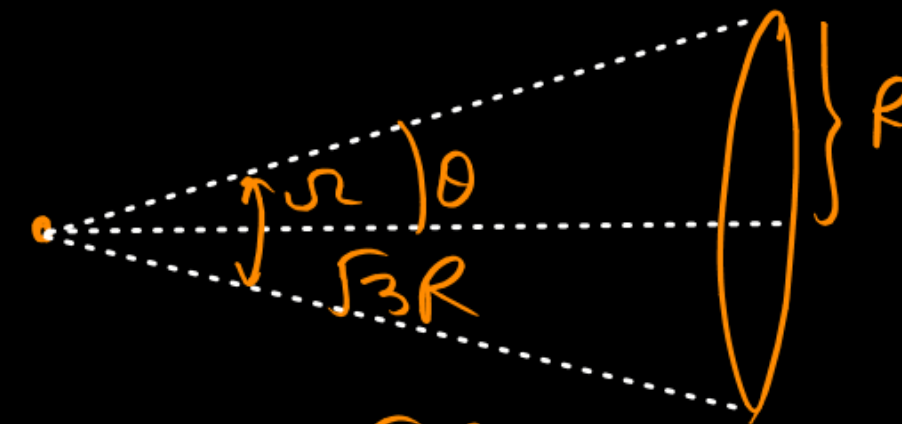
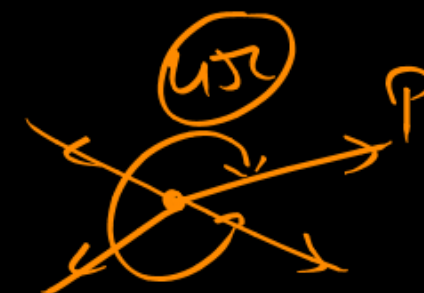


Figure -2



$$\Omega = 2\pi \left\{ 1 - \frac{\sqrt{3}R}{2R} \right\}$$

$$= 2\pi (2 - \sqrt{3}) = (2 - \sqrt{3})\pi$$

$$\Omega \rightarrow \frac{P}{4\pi} \Omega = \frac{P}{4\pi} \times \pi (2 - \sqrt{3})$$

$$P_1 = \frac{2 - \sqrt{3}}{4} P$$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P_1 . Now a transparent cylinder which does not absorb any light of a refractive index $\mu(= \sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

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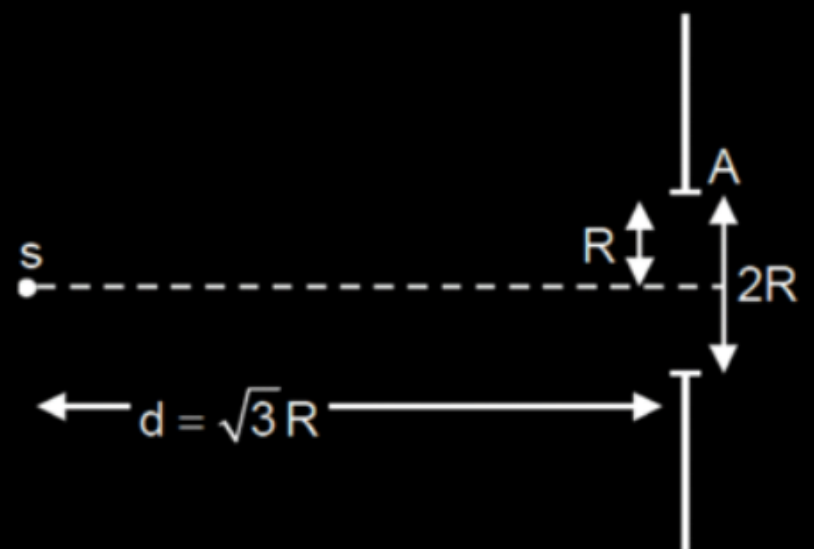


Figure -1

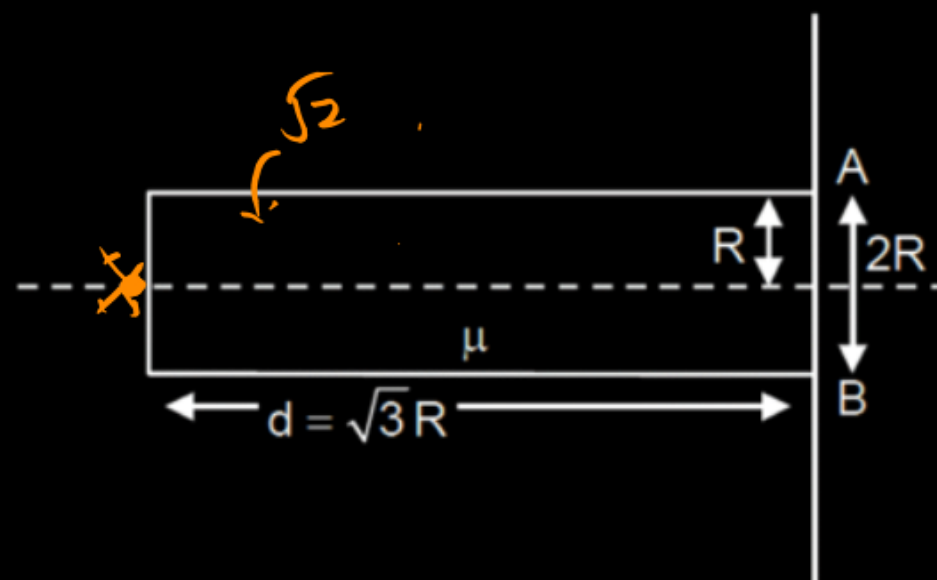


Figure -2

Answer the following two questions :

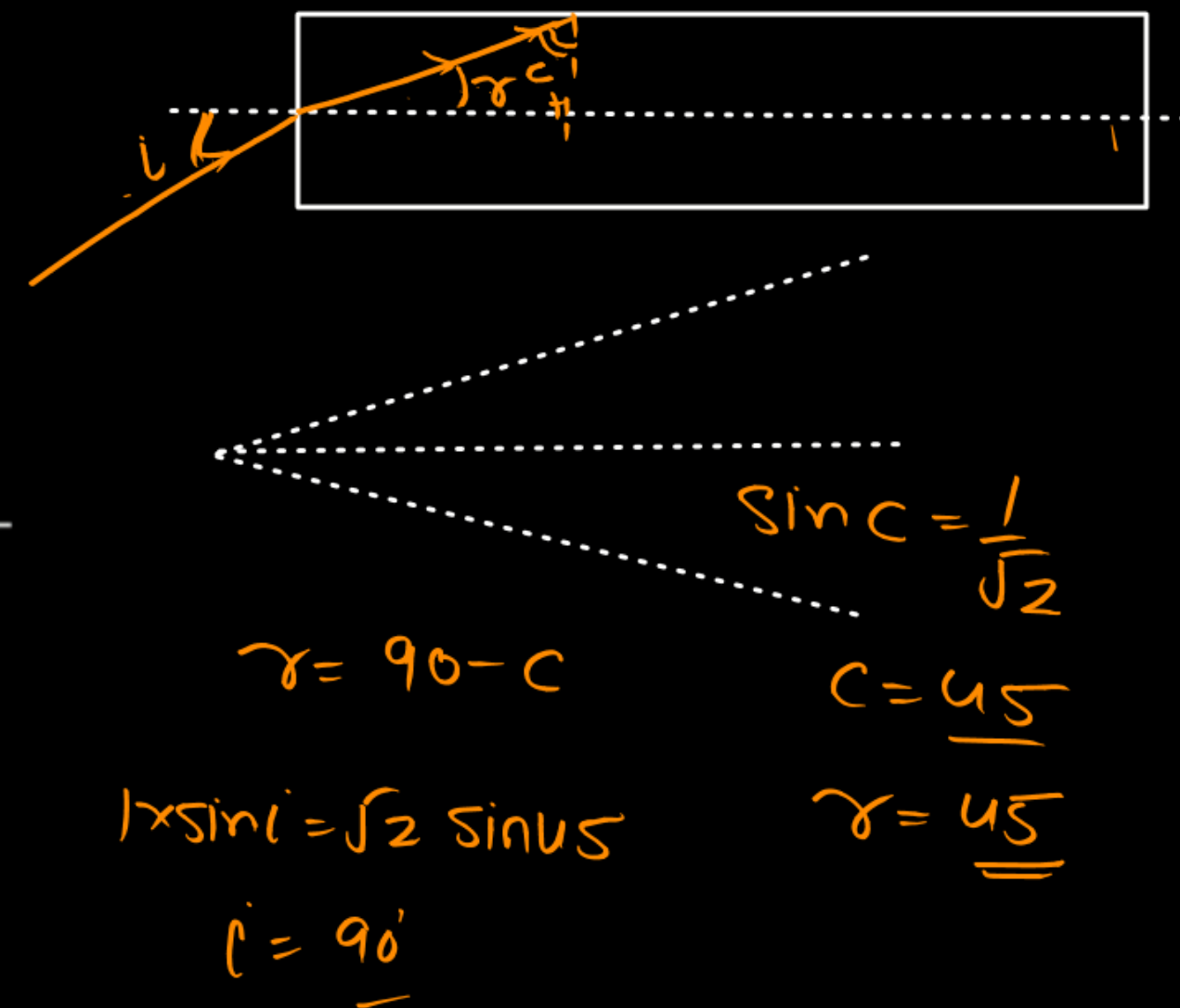
Value of P_2 is :

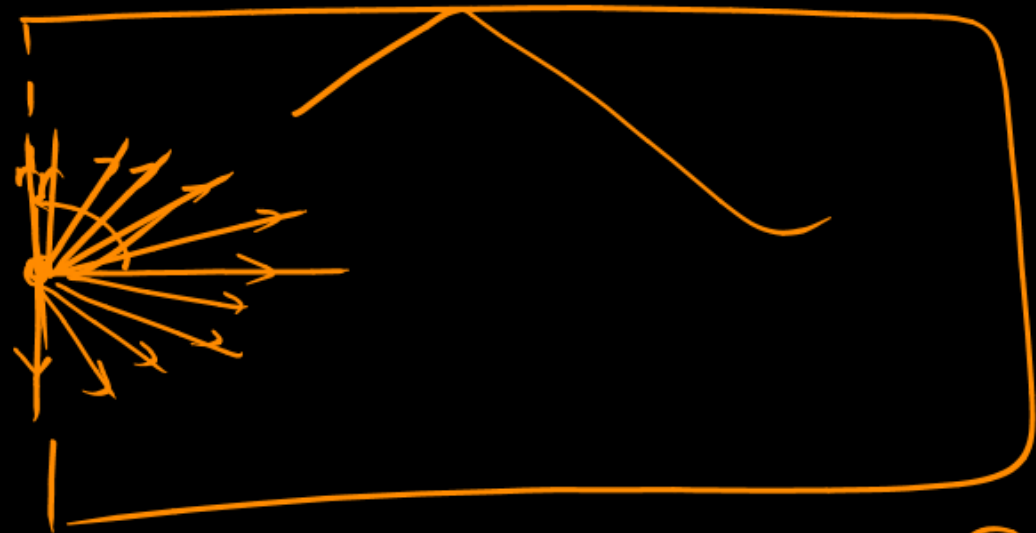
(A) $\frac{2P_1}{2 - \sqrt{3}}$

(B) $\frac{P_1}{2 - \sqrt{3}}$

(C) $\frac{2P_1}{2 + \sqrt{3}}$

(D) $\frac{P_1}{2 + \sqrt{3}}$





$$\left(\frac{P}{2} \right)$$

$$\begin{aligned} \Phi &= P_2 = \frac{P}{4\pi} \Omega = \frac{P}{4\pi} \times 2\pi (1 - \cos 90^\circ) \\ &= \frac{P}{2} = \frac{4P_1}{(2-\sqrt{3})^2} = \frac{2P_1}{(2-\sqrt{3})} \end{aligned}$$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P_1 . Now a transparent cylinder which does not absorb any light of a refractive index $\mu(=\sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

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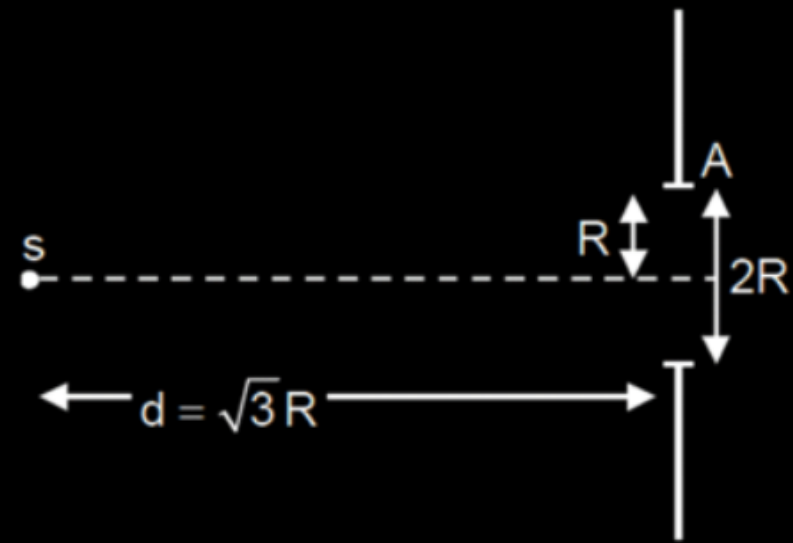


Figure -1

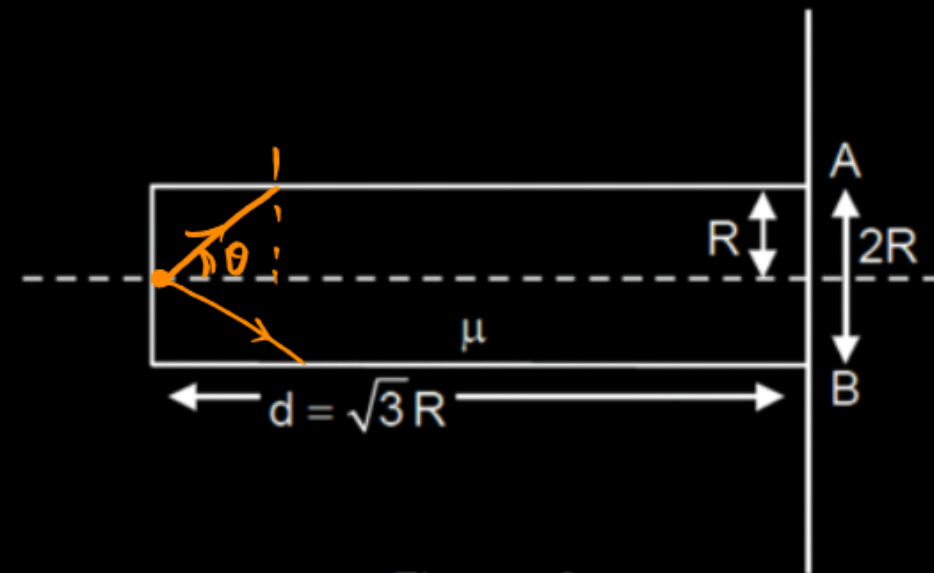


Figure -2

Answer the following two questions :

Value of P_3 is :

(A) $P_1 \left(\frac{2+\sqrt{2}}{2+\sqrt{3}} \right)$

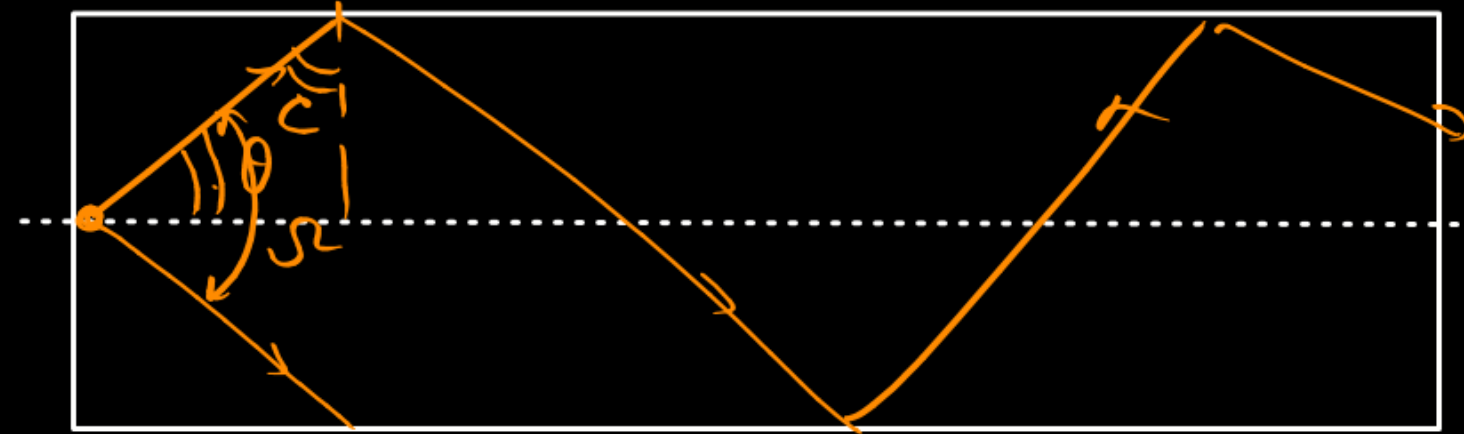
(B) $P_1 \left(\frac{2+\sqrt{2}}{2-\sqrt{3}} \right)$

(C) $P_1 \left(\frac{2-\sqrt{2}}{2+\sqrt{3}} \right)$

(D) $P_1 \left(\frac{2-\sqrt{2}}{2-\sqrt{3}} \right)$

$$P_3 = \frac{P}{4\pi} \times 2\pi \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{P}{2} \left(1 - \frac{1}{\sqrt{2}} \right) = \frac{4P_1}{(2-\sqrt{3}) \times 2} \frac{\sqrt{2}-1}{\sqrt{2}}$$



$$\theta = 90^\circ - C$$

$$= 90 - 45^\circ = 45^\circ$$

$$P_3 = \frac{2P_1(\sqrt{2}-1)}{\sqrt{2}(2-\sqrt{3})}$$

$$\mu = 2n(1 - \cos 45^\circ) \quad P_3 = \frac{(2-\sqrt{2})P_1}{(2-\sqrt{3})}$$

$$= 2n \left(1 - \frac{1}{\sqrt{2}} \right)$$

A radio consists of two isotopes. One of the isotopes decays by α -emission and the other by β -emission with half life $T_1 = 405$ second and $T_2 = 1620$ second, respectively, At $t = 0$, probabilities of getting α and β -particles from the radionuclide are equal.

If at $t = 0$, total number of nuclei in the radio nuclide are N_0 , calculate time $t/100$ (in sec.) when total number of nuclei remained undecayed becomes equal to $\frac{N_0}{2}$.

Given $\log_{10} 2 = 0.30103$ $\log_{10} 5.94 = 0.7742275$, $X^4 + 4x - 2.5 = 0 \Rightarrow x = 0.594$

$$\begin{array}{c} X \\ T_{\frac{1}{2}} = 405 = T \\ N_1 = \frac{N_0}{5} \end{array}$$

$$\begin{array}{c} Y \\ T_{\frac{1}{2}} = 1620 = 4T \\ N_2 = \frac{4N_0}{5} \end{array}$$

$$\begin{array}{l} t=0 \\ \lambda_1 N_1 = \lambda_2 N_2 \\ \frac{\ln 2}{T} N_1 = \frac{\ln 2}{4T} N_2 \\ \boxed{N_2 = 4N_1} \end{array}$$

$$N = N_1 + N_2$$

$$= \frac{N_0}{5} \left\{ 1 - 2^{-\frac{t}{T}} \right\} + \frac{4N_0}{5} \left\{ 1 - 2^{-\frac{t}{4T}} \right\} = \frac{N_0}{2}$$

$$2^{-\frac{t}{4T}} = x$$

$$2^{-\frac{t}{T}} = x^4$$

$$\frac{1}{5} + \frac{4}{5} - \frac{2}{5} - \frac{4}{5} 2^{-\frac{t}{4T}} = \frac{1}{2}$$

$$\frac{1}{5} 2^{-\frac{t}{T}} + \frac{4}{5} 2^{-\frac{t}{4T}} = \frac{1}{2}$$

$$x^4 + 4x = \frac{5}{2}$$

$$\boxed{x^4 + 4x - 2.5 = 0}$$

$$x = 0.594 = 2^{-\frac{t}{T}} = 2^{-\frac{t}{405}}$$

$$t = 1215 \text{ sec}$$

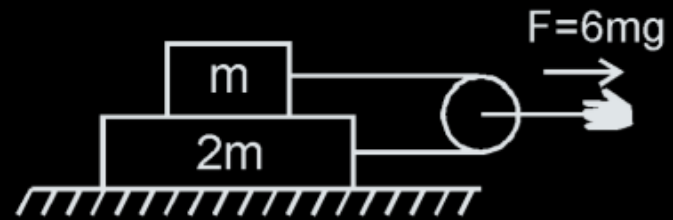
$$\boxed{\frac{t}{180} = 12.15}$$

A solid sphere of diameter 0.1 m is at 427°C and is kept in an enclosure at 27°C. Take Stefan's constant

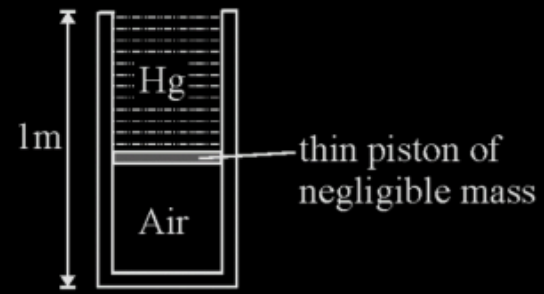
$= \frac{20}{3} \times 10^{-8} \text{ W/m}^2\text{K}^4$, emissivity of the surface 0.84, specific heat 0.1 kcal/kg K, density = 9280 kg/m³, J = 4200 J/k

cal. If rate of decrease of temperature of the sphere is $N \times 10^{-3}^\circ\text{C/s}$, find $\frac{N}{40}$.

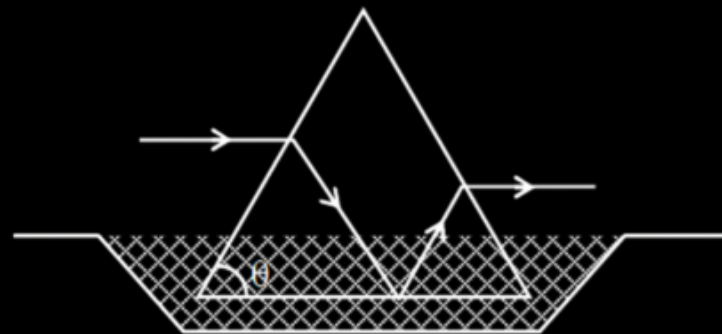
A block of mass m is placed on top of a block of mass $2m$ which in turn is placed on fixed horizontal surface. The coefficient of friction between all surfaces is $\mu = 1$. A massless string is connected to each mass and wraps halfway around a massless and frictionless pulley, as shown. The pulley is pulled by horizontal force of magnitude $F = 6mg$ towards right as shown. If the magnitude of acceleration of pulley is $\frac{X}{2} \text{ m/s}^2$, fill the value of X . (Take $g = 10 \text{ m/s}^2$)



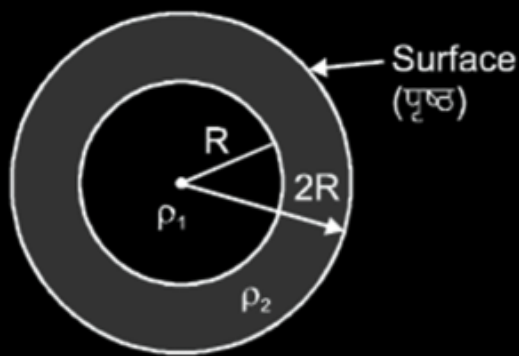
The 1 m tall conducting cylinder in figure contains air at a pressure of 76 cm of Hg. A very thin piston of negligible mass is placed at the top of the cylinder to prevent any air from escaping, then mercury is very slowly poured into the cylinder until no more can be added without the cylinder overflowing. What is the height h (in cm) of the compressed air? Fill $h/19$ in OMR



An isosceles triangular glass prism stands with its base in water as shown. A ray is incident parallel to base of prism. After refraction through inclined face it just suffers total internal reflection on water glass interface. If value of $\tan\theta$ is $\frac{\lambda}{\sqrt{17}}$ then find λ . (Take $\mu_{\text{glass}} = 3/2$, $\mu_{\text{water}} = 4/3$)



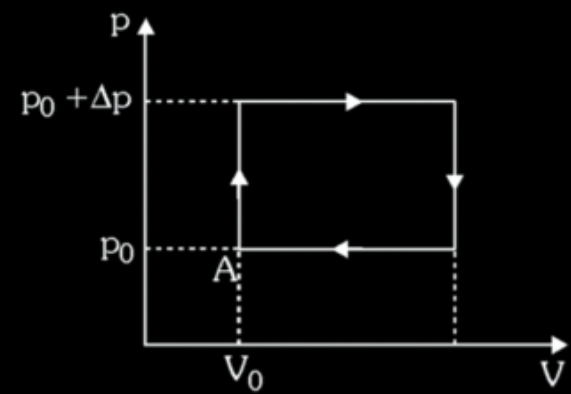
A planet is made of two materials of density ' ρ_1 ' and ' ρ_2 ' as shown in figure.



Acceleration due to gravity at surface of planet is same as that at depth 'R'. The ratio $\frac{\rho_1}{\rho_2}$ is equal to $\frac{x}{3}$.

Find the value of 'x'.

The working substance of a heat engine is a certain amount of monoatomic ideal gas. The gas is taken through the cyclic process shown in the figure which consists of two isobaric and two isochoric processes. Net work done by gas in a cycle is same as the internal energy of the gas in its initial state A. If for maximum possible efficiency of the cycle $\Delta p = np_0$, then find the value of $2n^2$.



A point charge $+q$ is placed on the axis of a closed cylinder of radius R and length $\frac{25R}{12}$ as shown. If

electric flux coming out from the curved surface of cylinder is $\frac{xq}{10\epsilon_0}$, then calculate x .



An ideal DC source of emf E is connected with an uncharged capacitor and inductor. If switch is closed at $t = 0$,

what is the maximum charge (in μC) on the capacitor in subsequent flow of charge ?

(Given : $L = 4\text{H}$; $C = 4.7\ \mu\text{F}$ and $E = 5.3\text{ volt}$)

